MILITARY SPECIFICATION

PAD, TRETINOIN SOLUTION IMPREGNATED, RAYON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Pad, Tretinoin Solution Impregnated, Rayon. (See 6.2.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Box, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.
PPP-C-00186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a rayon swab measuring approximately 1 inch by 1-1/4 inch saturated with approximately 1.4 ml of Tretinoin liquid containing 0.05 percent Retinoic Acid within the designated assay limits, 50 percent Specially Denatured Alcohol #40, 190 proof within the designated assay limits, and Polyethylene Glycol 400, and Butylated Hydroxytcluene.

3.2 Assay of the expressed solution from pad.

3.2.1 Retinoic acid. Shall contain not less than 0.045 percent and not more than 0.675 percent (W/W) when determined as specified in 4.4.1.

3.2.2 SDA alcohol #40. Shall contain not less than 50.0 percent and not more than 60.0 percent by volume, calculated on an absolute alcohol basis at 15.56°C, when determined as specified in 4.4.2.

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3.3 Liquid in pad. Each pad shall be saturated with 1.4 ml ± 0.15 ml equivalent to 1.33 grams ± 0.13 gram, when determined by a suitable method.

3.4 Bacterial count. The expressed solution from the pad shall contain not more than 10 viable micro-organisms per ml. None of the following organisms shall be present: Staphylococcus, Pseudomonas, Salmonella Species, and Escherichia coli. Presence or absence of micro-organisms shall be determined by appropriate, accurate, reproducible method(s).

3.5 Ingredients.

3.5.1 Retinoic acid.

3.5.1.1 Description. Shall be a yellow to light orange crystalline powder.

3.5.1.2 Infrared identification. Shall comply with 4. .1.1.

3.5.1.3 Ultraviolet spectrum. Shall comply with 4. .1.2.

3.5.1.4 Loss on drying. Shall not exceed 0.5 percent when determined as specified in 4. 5.1.3.

3.5.1.5 Residue on ignition. Shall not exceed 0.1 percent when determined as specified in 4. 5.1.4.

3.5.1.6 Heavy metals. Shall not exceed 20 parts per million when determined as specified in 4. 5.1.5.

3.5.1.7 Assay. Shall assay to contain not less than 97.0 percent and not more than 103.0 percent retinoic acid when determined as specified in 4. 5.1.6.

3.5.2 Butylated hydroxytoluene.

3.5.2.1 Description. Shall be white granular crystals with a very slight odor.

3.5.2.2 Melting range. Shall melt between 69° and 71°C. when determined as specified in 4. 5.2.1.

3.5.2.3 Infrared identification. Shall comply with 4. 5.2.2.

3.5.2.4 Residue on ignition. Shall not exceed 0.1 percent when determined as specified in 4. 5.2.3.

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3.5.3 SDA Alcohol No. 40 (190 proof).

3.5.3.1 Description. Clear, colorless liquid, volatile, with a slight characteristic odor.

3.5.3.2 Specific gravity. Shall contain not less than 94.9 percent by volume equivalent to a specific gravity not more than 0.8096 at 25/25°C. when determined as specified in 4.5.3.1.

3.5.3.3 Ultraviolet identification. Shall comply with 4.5.3.2.

3.5.4 Polyethylene Glycol 400. Shall be in accordance with the tests, standards and requirements of the U.S.P. including any supplements, or revisions, thereto.

3.6 Other ingredients. All other ingredients entering into the manufacture of the solution shall be of U.S.P. or N.F. quality or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical quality and suitable for use in the manufacture of the impregnated pad.

3.7 Workmanship. Material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

3.8 Delivery schedule. Not less than 33 months of the expiration dating period (potency period) shall remain at the time of delivery to the Government.

3.9 If (i) the quality assurance representative submits samples to the laboratory at the Defense Personnel Support Center, or (ii) the bidder or offeror is required by the terms of the Schedule or otherwise to submit samples to said Laboratory, a quantity of 5 grams of Retinoic Acid used in the manufacture of each lot of Tretinoin solution and an infrared spectrum of the Reference Standard, accompanied by a certificate of analysis listing all test results, shall be submitted with the finished product to the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, Attention: Quality Assurance Branch, 2800 South 20th Street, Philadelphia, PA 19101. Samples shall be packaged in dry ice during the period of shipment to Defense Personnel Support Center.

3.10 Potency period (Expiration dating period). Potency period (Expiration dating period) shall be 36 months.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purpose of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of finished solution produced from one formulation, subjected to the same compounding and manufacturing operations, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with the procedure set forth in Military Standard MIL-STD-105. With an acceptable quality level (AQL) of 1.5 percent defective for major defects and 4.0 percent defective for minor defects, using Inspection Level S-3.

4.4 Assay of the finished solution expressed from the finished pad.

4.4.1 Retinoic acid. (Caution: Since retinoic acid is a skin irritant, use adequate eye protection and avoid skin contact. If any of the material comes in contact with the skin wash well with soap and water.) Retinoic acid decomposes in the presence of light. Perform the analysis without undue delay while protecting samples and standards from light. Retinoic acid standards should be stored in a suitable apparatus which maintains a constant temperature of -10°C + -3°C. Standard must be assayed in accordance with 4.5.1.6 in order to determine exact percentage retinoic acid to be used in calculation.

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Standard solution

Accurately weigh approximately 75 mg of retinoic reference standard into a 100-ml amber volumetric flask. Dissolve in and dilute to volume with isopropanol. Pipet 10.0 ml of the above solution into a second 100-ml amber volumetric flask and dilute to volume with isopropanol. Pipet 4.0 ml of the solution into a third 100-ml amber volumetric flask and dilute to volume with isopropanol.

Procedure

Express the liquid from 20 pads into a container protected from light and having a closure to prevent evaporation. Accurately weigh approximately 600 mg of finished solution into an amber 100-ml volumetric flask and dilute to volume with isopropanol. Concomitantly determine the absorbances of sample and standard solutions at the wavelength of maximum absorbance, at approximately 350 mu with a suitable spectrophotometer in a 1-cm cell using isopropanol as the blank. Calculate the percentage W/W of retinoic acid in the finished sample.

4.4.2 SDA alcohol #40. Shall be determined in accordance with a suitable, accurate, and reproducible gas chromatographic method.

4.5 Ingredients.

4.5.1 Retinoic acid.

4.5.1.1 Infrared spectrum. The infrared absorption spectrum of a mineral oil mull containing at least 5 percent retinoic acid, from 2-15 microns, exhibits maxima only at the same wavelengths as that of a similarly prepared Reference Standard. A copy of the IR spectrum of the Reference Standard shall be submitted by the contractor and shall be kept on file in the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, for use if samples are submitted.

4.5.1.2 Ultraviolet spectrum. The ultraviolet spectrum of a solution of a final concentration of approximately 3 mcg per ml in isopropanol exhibits maxima at the same wavelengths and conforms to that of a similarly prepared solution of Retinoic Acid Reference Standard. A copy of the UV spectrum of the Reference Standard shall be submitted by the contractor and shall be kept on file in the Technical and Quality Assurance Division, Defense Personnel Support Center.

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4.5.1.3 Loss on drying. Shall be determined by the U.S.P. procedure drying the sample for 3 hours at 105°C.

4.5.1.4 Residue on ignition. Shall be determined in accordance with the U.S.P. method.

4.5.1.5 Heavy metals. Shall be determined in accordance with U.S.P. method II, or suitably modified as required.

4.5.1.6 Assay. (NOTE: During titration protect solution from water and atmospheric carbon dioxide.) Accurately weigh approximately 240 mg of sample into an Erlenmeyer flask. Add 50 ml of N, N-dimethylformamide, and several drops of 1 percent thymol blue in N, N-dimethylformamide indicator solution. Titrate with 0.1N sodium methoxide U.S.P. Volumetric Solution to a blue end point which persists for 15 seconds. (Since solution is yellow end point may be slightly greenish.) Perform a blank determination and make any necessary correction. Each ml of 0.1N sodium methoxide solution is equivalent to 30.04 mg of retinoic acid.

4.5.2 Butylated hydroxytoluene.

4.5.2.1 Melting range. Shall be determined by the U.S.P. Method for Class I.

4.5.2.2 Infrared identification. Place a small portion of sample on an optical plate and carefully melt sample. Place another optical plate over the sample, and rotate gently in order to affect uniform spreading. The infrared absorption spectrum exhibits maxima only at the same wavelengths as that of the Reference Standard. A copy of the IR spectrum of the Reference Standard shall be submitted by the contractor and shall be kept on file in the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, for use if samples are submitted.

4.5.2.3 Residue on ignition. Shall be determined in accordance with the U.S.P. method.

4.5.3 SDA alcohol No. 40.

4.5.3.1 Specific gravity. Shall be determined in accordance with the U.S.P. pycnometer method.

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4.5.3.2 Ultraviolet identification. The ultraviolet spectrum between 200 to 400 mu exhibits maxima at the same wavelengths and conforms to that of a similarly prepared solution of Brucine Sulfate, NF IX, which was official from 1 November 1950 to 15 December 1955 (or brucine alkaloid) in 95 percent U.S.P. alcohol. A copy of the UV spectrum of the Reference Standard shall be submitted by the contractor and shall be kept on file in the Technical and Quality Assurance Division, Directorate of Medical Material, Defense Personnel Support Center.

4.6 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table 1. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table I. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Preparation for Delivery shall be in accordance with all applicable requirements of PPP-C-00186 and as specified herein:

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One box containing thirty individually packaged pads, as specified, constitutes one unit of issue.

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5.2.2 Level A.

5.2.2.1 Immediate container. Immediate container shall be a commercially available container considered acceptable to the Food and Drug Administration (FDA) for the product contained therein. The container shall be of such compatibility that neither the contents nor the container shall be altered in any way by each other.

5.2.2.2 Unit package. Thirty immediate containers shall be packaged in a snug-fitting box constructed in accordance with PPP-B-566 or PPP-B-676. Closure of the box shall be secured.

5.2.2.3 Intermediate package. Twelve unit packages shall be packaged in a snug-fitting box constructed in accordance with PPP-B-566, PPP-B-636, type CF, class domestic or PPP-B-676. Closure of the box shall be secured.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.3.1 Level A. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant except that grade W5c is not acceptable as an exterior container. Closure, waterproofing and strapping shall be as specified in the appendix of the box specification.

5.3.2 Level B. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic and having a minimum bursting strength of 200 pounds. Closure shall conform to method II as specified in PPP-B-636.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.4 Marking.

5.4.1 Labeling shall be in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act and shall bear the following information:

5.4.1.1 Immediate container. Each immediate container shall bear the following information:

(a) labeling information in accordance with commercial practice.

5.4.1.2 Unit package. Each unit package shall bear the following information:

(a) labeling in accordance with commercial practice.
(b) the National Stock Number.
(c) the expiration date.

NOTE: A strip label may be used to indicate the National Stock Number.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number and name of contractor shall be shown. Type I Shelf Life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F)".

5.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. Type I Shelf Life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

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6.2 This specification covers the following item(s):

National Stock Number	Item Identification
6510-00-117-9010	PAD, TRETINOIN SOLUTION IMPREGNATED, RAYON, 1-1/4 by 1 inch, 30s

6.3 Supersession data. This specification includes the requirements of Defense Medical Purchase Description No. 1 dated 30 October 1973.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6510-0607

Review information is current as of date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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